

Andrii Nikolaiev, PhD

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SUMMARY

AI Researcher with a PhD in Computer Science and 3+ years of NLP research at the University of Cambridge. 10+ years in EdTech, from lecturer to building national STEM programmes. Driven by a passion for using technology to make learning more effective.

EDUCATION

- Visiting Research Student**
NLIP Group, University of Cambridge (2023–2025)
- Ph.D. in Computer Science**
Taras Shevchenko National Univ. of Kyiv (2020–2025)
- B.Sc. & M.Sc. in Computer Science**
Taras Shevchenko National Univ. of Kyiv (2014–2020)

GRANTS & AWARDS

- Emergent Ventures Fellow, Mercatus Center, USA (2023–2026)
- Fully funded research fellowship, University of Cambridge, UK (2023–2025)

SKILLS

- LLM Evaluation & Research:** Benchmarking, CoT, LLM-as-a-Judge, Prompt Engineering, Human Evaluation, Rubric Design
- Programming languages:** Python, SQL, R, C/C++, $\LaTeX$
- ML Frameworks & Inference:** PyTorch, Hugging Face, TensorFlow, vLLM, llama.cpp, Ollama, CUDA
- Infrastructure:** HPC, Google Cloud Platform, Git, Linux/UNIX, Kubernetes, Docker
- Natural languages:** English (fluent), Ukrainian (native)
- Interests:** Half-marathons (PB: 01:31:16), football, hiking, yoga

RESEARCH & PROFESSIONAL EXPERIENCE

- Project Manager & Researcher, Mercor** Feb 2026 – Present
- Designed pilot studies, evaluation rubrics and experiments assessing frontier AI models on economically meaningful tasks (*APEX-Agents*, *Terminal-Bench*).
 - Analysed AI agentic behaviour – MCP tool usage, computer & browser user actions (CUA, BUA), reasoning, code execution.
 - Oversaw teams of 700+ people across several project directions; collaborated with stakeholders and domain experts across industries.

- Researcher in NLP, University of Cambridge** Apr 2023 – Present
- Developed *Combi-Puzzles*, a novel symbolic benchmark for combinatorial reasoning, revealing reasoning gaps across frontier LLMs.
 - Deployed and evaluated open-source and proprietary LLMs on HPC clusters and cloud VMs.
 - Designed and led human evaluation studies, comparing model responses against expert judgements.

- Researcher in LLM Benchmarking, Osiris AI** Aug – Sep 2025
- Architected a distributed multimodal pipeline for OCR and segmentation of historical handwritten records via custom RAG methods.

- Ph.D. in Computer Science, Taras Shevchenko National Univ.** 2020 – 2025
- Thesis on LLM evaluation for abstract thinking and problem-solving, revealing differences in reasoning patterns between models and humans.
 - Conducted large-scale benchmarking and analysis of LLM reasoning using over 1M Chain-of-Thought examples.
 - Implemented LLM-as-a-Judge frameworks, reducing evaluation costs while maintaining strong alignment with expert rubrics.

SELECTED PUBLICATIONS

- Neural Network Methods for Selecting and Generating Synthetic Variations of Combinatorial Problems, *Cybernetics and Systems Analysis*, Springer Nature, 2025.
- Can Language Models Rival Mathematics Students? Evaluating Mathematical Reasoning through Textual Manipulation and Human Experiments, *ACL RR*, arXiv preprint, 2024.

LEADERSHIP EXPERIENCE

- Co-founder & CEO, Kvanta.STEM EdTech** 2016 – Present
- Built a national year-round STEM programme with 1,500+ learners and 70+ instructors, securing \$250K+ in grants and sponsorship for its educational impact from *Emergent Ventures*, *XTX Markets*, *GlobalLogic*, and *Admixer*.
 - Designed Olympiad-level programs in mathematics, programming, and science; organised boot camps for up to 300+ participants (online & in-person).

Mentorship & Other Roles

- Instructor and lecturer at STEM-focused schools and boot camps, including *Mathematical Olymp*, *School of Programming and Information Technology*, *Ukrainian Physics and Mathematics Lyceum*.
- Managed national on-site educational programmes at *NGO Kontora Pi*.
- Assisted *Databases* and *Cloud Computing* at *KNU*; *Machine Learning for Real-World Data* seminars at the *University of Cambridge*.